

# Non-isometric transforms in time series classification using DTW



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## ABSTRACT

Over recent years the popularity of time series has soared. As a consequence there has been a dramatic increase in the amount of interest in querying and mining such data. In particular, many new distance measures between time series have been introduced. In this paper we propose a new distance function based on derivatives and transforms of time series. In contrast to well-known measures from the literature, our approach combines three distances: DTW distance between time series, DTW distance between derivatives of time series, and DTW distance between transforms of time series. The new distance is used in classification with the nearest neighbor rule. In order to provide a comprehensive comparison, we conducted a set of experiments, testing effectiveness on 47 time series data sets from a wide variety of application domains. Our experiments show that this new method provides a significantly more accurate classification on the examined data sets.

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## 1. Introduction

In recent decades, time series analysis has become one of the most popular branches of statistics. Time series are currently ubiquitous, and have come to be used in many fields of science. Data sets in the form of time series occur in many areas of human life. One of the major applications is time series classification. The problem of time series classification has been intensively investigated during the last few years, with the increase in the computational capabilities of computers. It is particularly important to be able to find the distance between time series. We can find many distance measures for similarity of time series data. Nevertheless, the simple method combining the nearest neighbor (1NN) classifier and some form of dynamic time warping (DTW) distance has been shown to be one of the best-performing time series classification techniques [8]. DTW is a classical and well-established distance measure well suited to the task of comparing time series [6].

It seems that in the classification domain there could be objects for which function value comparison is not sufficient. There could be cases where assignment to one of the classes depends on the general shape of objects rather than on strict function value comparison. In mathematics, an object associated with a function that responds to its variability in “time” is the derivative of the function. We cannot expect that will be sufficient to compare only time

series derivatives. It seems that the best approach is to create a method which considers both function values of time series and values of the derivative of the function. The strength of the influence of these approaches should be parameterized. In this way we are able to deal with situations where the examined sequences are not different enough. For a distance function a new parameterized family of distance measures is formed, where the fixed distance measure is used to compute distances of time series and their variability in “time”.

The use of derivatives in time series classification is not a novelty. Their use with DTW was proposed by Keogh and Pazzani [15], who called their method Derivative Dynamic Time Warping (DDTW). On the basis of this method, Kulbacki et al. [18] created a measure which took into account the point-to-point distance between time series. The authors named their method Value-Derivative Dynamic Time Warping (VDDTW). Benedikt et al. [3,4] have gone even further. They introduced a weighted sum, in which they also took into consideration the second derivative. They called their method Weighted Derivative Dynamic Time Warping (WDDTW). Our previous work [12] contains the results of research on DDTW where the derivative is added, while at the same time parameterization involves both function and derivative. As was shown, our method outperforms classical DDTW, VDDTW and WDDTW. For this reason, in the present work we decided to compare our new method, involving the transform, only with DD<sub>DTW</sub>.

Because the addition of first derivative gave such good results, we decided to conduct further research. We were looking for functions other than the derivative which can be used in a similar manner.

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The choice was mathematical transforms, which are very popularly used in the classification of time series. However, they have so far been used mainly for representation, allowing efficient computation on the data and extraction of higher order features [9]. For this purpose one needs distance-preserving transforms like the Fourier transform. The Fourier transform is a one-to-one linear transformation from the linear space  $L^2(-\infty, \infty)$  to the linear space  $L^2(-\infty, \infty)$  i.e. the Fourier transform is an isomorphism [11]. However, we want to use the transform to calculate distances, and add it as another component of the measure of distance between time series. In this case we use the DTW distance measure (instead of Euclidean distance), so the Fourier transform is non-isometric. If the transform is an isometry, it does not bring any new information as to the distance between the time series, because it is a distance-preserving map between metric spaces (they preserve lengths and inner products). Ultimately, we decided on three real transforms: sine, cosine and Hilbert. Of course, as before, we do not think that it is enough to compare only the distance between the transforms. It seems natural to add transforms as a further element to improve the accuracy of classification. As before, the parametric approach was used, which allows us to choose the impact of each distance on the final distance measure between the time series, and consequently on the quality of the classification. The new distance functions so constructed are used in the nearest neighbor classification method.

The method works well on a wide range of data sets. In spite of the need to search for a large subset of parameters, the computational complexity is actually like for a method with one parameter. For all these reasons in combination, our method seems to be a universal method for the classification of time series, able to identify for which of the data sets the impact of the derivatives is helpful and to what extent.

The remainder of the paper is organized as follows. We first (Section 2) review the concept of time series and the dynamic time warping algorithm, and we introduce transforms. In Section 3 we introduce our distance, combining the distances between time series and transforms. In the same section we explain the optimization process and properties of our new distance measure in this 2D case. We go on to describe the data sets and experimental setup and present the results of our experiment. Section 4 is similar to Section 3, except that in this case we examine a distance measure combining distances between time series, derivatives and transforms. We conclude in Section 5 with discussion of possible future extensions of the work.

## 2. Basic methods

A time series is a sequence of observations which are ordered in time or space [7]. For simplicity and without any loss of generality, we assume that time is discrete. Formally, a time series data is defined as a sequence of pairs  $T = [(\mathbf{x}_1, t_1), (\mathbf{x}_2, t_2), \dots, (\mathbf{x}_n, t_n)]$  ( $t_1 < t_2 < \dots < t_n$ ) where each  $\mathbf{x}_i$  is a data point in  $d$ -dimensional feature space, and each  $t_i$  is the time stamp at which  $\mathbf{x}_i$  occurs. If the sampling rates of two time series are the same, we can omit the time stamps and consider them as sequences of  $d$ -dimensional data points. Such a representation is called the raw representation of the time series. The number of data points  $n$  in a given time series is called its length. In the rest of this paper we will use only one-dimensional time series, which are conveniently represented in the form  $x(i)$ ,  $i = 1, 2, \dots, n$ .

### 2.1. Dynamic time warping

Dynamic time warping is a well-known technique to find an optimal alignment between two given time-dependent sequences under certain restrictions. Originally, DTW was used to compare different speech patterns in automatic speech recognition. It is an

algorithm for measuring similarity between two sequences which may vary in time or speed. The sequences are warped in a nonlinear fashion to match each other. If we have two signals encoded as a sequence of evenly spaced values, then an obvious way to compare the signals is to sum the differences in frequency at each point along the signals. However, a problem arises if there is any discrepancy in the alignment of the signals.

To find DTW distance, we first construct a matrix where the  $(i, j)$ th element corresponds to  $d(x_i, y_j) = (x_i - y_j)^2$ , which is the alignment between points  $x_i$  and  $y_j$ . Then we find a path through the matrix with minimal cumulative distance between time series (Fig. 1). DTW distance corresponds to the path with minimal warping cost:

$$DTW(x, y) = \min \sqrt{\sum_{k=1}^K w_k},$$

where  $w_k$  is the matrix element that also belongs to the  $k$ th element of a warping path  $W$ . The warping path is subject to three constraints:

- $w_1 = (1, 1)$  and  $w_K = (n, m)$  (boundary conditions);
- For  $w_k = (a, b)$  and  $w_{k-1} = (a', b')$ ,  $a - a' \leq 1$  and  $b - b' \leq 1$  (continuity);
- For  $w_k = (a, b)$  and  $w_{k-1} = (a', b')$ ,  $a - a' \geq 0$  and  $b - b' \geq 0$  (monotonicity).

We can use dynamic programming to find this warping path, applying the following recurrence:

$$\gamma(i, j) = d(x_i, y_j) + \min\{\gamma(i-1, j-1), \gamma(i-1, j), \gamma(i, j-1)\},$$

where  $d(x_i, y_j)$  is the distance found in the current cell, and  $\gamma(i, j)$  is the cumulative distance of  $d(x_i, y_j)$  and the minimum cumulative distances from the three adjacent cells.

### 2.2. Transforms

We shall now give the definitions of the transforms used in this paper. They are three transforms that are very popularly used in the technical sciences: the cosine transform, sine transform, and Hilbert transform. All three are non-isometric for the DTW distance measure. For a series  $f = \{f(i) : i = 1, 2, \dots, n\}$  we have a transform  $\hat{f} = \{\hat{f}(k) : k = 1, 2, \dots, n\}$ .

Cosine transform:

$$\hat{f}(k) = \sum_{i=1}^n f(i) \cos \left[ \frac{\pi}{n} \left( i - \frac{1}{2} \right) (k-1) \right].$$

Sine transform:

$$\hat{f}(k) = \sum_{i=1}^n f(i) \sin \left[ \frac{\pi}{n} \left( i - \frac{1}{2} \right) k \right].$$

Hilbert transform:

$$\hat{f}(k) = \sum_{\substack{i=1 \\ i \neq k}}^n \frac{f(i)}{k-i}.$$

## 3. 2D methods

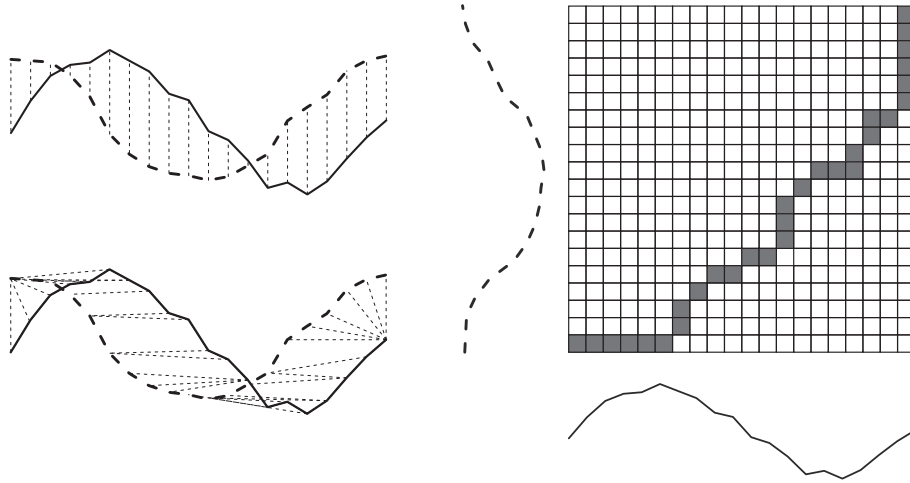
### 3.1. Distance measure

If  $\text{dist}$  is a distance measure for two time series  $f$  and  $g$ , a new distance measure  $\text{dist}_{ab}$  is defined by

$$\widehat{\text{dist}}_{ab}(f, g) := a \text{dist}(f, g) + b \text{dist}(\hat{f}, \hat{g}), \quad (1)$$

where  $\hat{f}$ ,  $\hat{g}$  are transforms of  $f$ ,  $g$ , and  $a, b \in [0, 1]$  are parameters.

To make sense of the definition, the transform  $\hat{\cdot}$  should not be an isometry for the distance  $\text{dist}$ . In this work, we use DTW distance with the sine, cosine, and Hilbert transforms.



**Fig. 1.** Top left: Two time series which are similar but out of phase produce a large Euclidean distance. Bottom left: This can be corrected by DTW's nonlinear alignment. Right: To align the signals we construct a warping matrix, and search for the optimal warping path.

The distance function  $\widehat{\text{dist}}_{ab}$  is used with the 1NN method in the classification process. The parameters  $a$ ,  $b$  are chosen in the learning phase. In this paper we use the cross-validation (leave-one-out) method on the learning data set in the process of parameter tuning.

Note that there is no reason why the distances between time series and between the transforms in (1) should not be different; for example, DTW between the transforms and Euclidean distance between the time series. However in this paper we always take the same distance function: DTW. For an arbitrary distance function  $\text{dist}$  and a transform  $\text{trans}$ , we will denote the new distance measure by  $\text{TD}_{\text{dist}}^{\text{trans}}$  (transform distance), for example  $\text{TD}_{\text{DTW}}^{\text{C}}$  (cosine transform).

### 3.2. Parameter dimension

We do not have to check all values of  $a, b \in [0, 1]$ . If  $a_1 = ca_2$  and  $b_1 = cb_2$ , where  $c > 0$  is a constant (i.e. points  $(a_1, b_1), (a_2, b_2)$  are linearly dependent), we have

$$\widehat{\text{dist}}_{a_1 b_1}(f_1, g_1) < \widehat{\text{dist}}_{a_1 b_1}(f_2, g_2) \iff \widehat{\text{dist}}_{a_2 b_2}(f_1, g_1) < \widehat{\text{dist}}_{a_2 b_2}(f_2, g_2)$$

so we can choose points  $(a, b)$  on any continuous line between the points  $(0, 1)$  and  $(1, 0)$ . We take the simplest case – the straight line (Fig. 2) with equation:

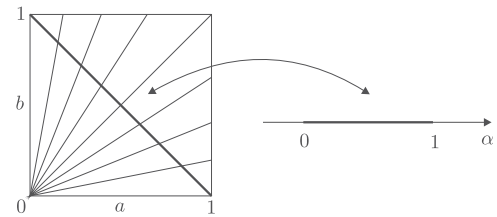
$$\begin{aligned} a &= (1 - \alpha), \\ b &= \alpha, \end{aligned} \quad \alpha \in [0, 1].$$

If the subset of the parameters  $\alpha$  is dense enough, the choice of parameterization should not be critical. In the next part of the paper we will use one parameter  $\alpha$  instead of  $a, b$ .

### 3.3. Parameter tuning

In the training phase we have to tune the parameter  $\alpha$ . Let  $A \subset [0, 1]$  be a finite subset of  $k$  parameters. We have to compute the cross-validation (leave-one-out) error rate on the learning data set for every  $\alpha \in A$  and choose the parameter for the smallest value of the error (1NN method). However, we can do this in several ways, differing in computational and memory complexity.

In standard procedures, the tuning of the parameter is as follows. We fix a parameter  $\alpha$  and then (by cross-validation) for every element of a training data set we calculate an appropriate number  $(n - 1)$  of distance functions  $\text{dist}$ . We repeat this for every



**Fig. 2.** Dependence of parameters  $a$ ,  $b$  and  $\alpha$ .

parameter  $\alpha$ . However, for our distance function, we can proceed in the opposite direction.

First, we fix one element  $e$  of the training data set. Then we take an element  $e_1 \neq e$ , and for every parameter  $\alpha$  we compute the distance function  $d = \widehat{\text{dist}}_{ab}(e_1, e)$  and put its value in a vector  $D$  (with  $k$  elements). Note that we need to compute only two distance measures  $\text{dist}$  (for the function and its transform). Then for the next element  $e_2 \neq e$  from the training set, we repeat the procedure and obtain a new distance vector  $d_2$ . Now we compare the elements in corresponding positions of the vectors  $D$  and  $d_2$ . In every position of vector  $D$  we insert the smaller one. We create a vector  $L$  (with  $k$  elements) which we fill with labels of elements  $e_1, e_2$  corresponding to values of the vector  $D$ . We repeat the procedure for the next elements of the training set ( $e_i \neq e, i = 3, 4, \dots, n$ ). As a result we obtain a vector  $L$  with labels of the nearest neighbors of the element  $e$  for every parameter  $\alpha$ . This is one step of the cross-validation process. We performed the classification of element  $e$  for all parameters  $\alpha$ . Note that we computed the distance function  $\text{dist}$  only  $2(n - 1)$  times. Repeating the procedure for all elements from the training data set, we obtain the cross-validation error rate for all parameters. The code of the algorithm is given in Section 3.5.

The computation of the distance  $\text{dist}$  has a complexity of  $O(n^2)$ . Because calculations of the distance functions  $\text{dist}$  are usually the most time-consuming part of the algorithm, the computation time depends to a small degree on the number of parameters (especially for large values of  $n$ ). Due to this fact, we can choose a large subset of parameters in the cross-validation process without increasing the computation time of the parameter tuning phase.

### 3.4. Lower bound and triangle inequality

To speed up the calculation we can use an approximation function called a lower bound. A lower bound is computationally much cheaper than the full DTW function. There are many distance

measures for which a lower bound can be found [16]. Two good lower bounds for DTW are LB\_Improved [19] and LB\_Keogh [16]. It is easy to find a lower bound for our new distance measure. If  $low$  is a lower bound for a distance  $dist$ , then

$$\widehat{low}_{ab}(f, g) := a \cdot low(f, g) + b \cdot low(\hat{f}, \hat{g})$$

is a lower bound for the distance  $dist_{ab}$ .

If the distance  $dist$  is a metric, then the new distance  $\widehat{dist}$  is also a metric. If  $dist$  is not a metric but obeys the triangle inequality, then the distance  $dist$  obeys the triangle inequality as well:

$$\widehat{dist}_{ab}(f, g) \leq \widehat{dist}_{ab}(f, h) + \widehat{dist}_{ab}(h, g).$$

Notice that DTW is not a metric and it does not satisfy the triangle inequality.

### 3.5. Experimental setup

We performed experiments on 47 data sets. Information on the data sets used is presented in Table 1. The data sets originate from the UCR Time Series Classification/Clustering Homepage [17].

For use with the new distance function, we chose one distance measure, DTW, and three transforms: sine, cosine, and Hilbert. Note, however, that our method can work with any distance measure and any non-isometric transforms. Thus we have three similarity measures denoted by  $TD_{DTW}^S$ ,  $TD_{DTW}^C$ , and  $TD_{DTW}^H$ . For each data set we calculated the classification error rate on a test subset (to learn the model we used a training subset with the leave-one-out, 1NN method). We found all parameters using the training subset. An appropriate distribution of the training and test sets is proposed by the authors of the repository (each data set is divided into a training and testing subset).

We use the cross-validation (leave-one-out) method to find the best parameter  $\alpha$  in our classifier. If the minimal error rate is the same for more than one value of  $\alpha$  we choose the smallest one. We implemented the algorithm discussed in Section 3.3. A finite subset of parameters  $\alpha$  is chosen, from 0 to 1 with fixed step 0.01. The code of the algorithm in MATLAB is presented below.

```
% e - list of time series in learning data set (cell
%       vector of vectors)
% labels - vector of labels of elements of list e
% dist - distance function (DTW)
% trans - transform function (sine, cosine, Hilbert)

step = 0.01;
alpha = 0 : step : 1;
a = 1 - alpha;
b = alpha;

n = length(e);
k = length(alpha);
mistakes(1:k) = 0; % vector of numbers of
%       misclassified elements

for i = 1 : n
    D(1:k) = inf; % vector of minimal distances
    L(1:k) = 0; % vector of 'minimal' labels
    for j = [1:i-1, i+1:n] % leave-one-out
        d = a * dist(e{j}, e{i}) + b * dist(trans(e{j}),
        trans(e{i}));
        D(d < D) = d(d < D);
        L(d < D) = labels(j);
    end
    mistakes = mistakes + (L == labels(i));
end
errors = mistakes/n; % error rates for every parameter
alpha
```

### 3.6. Results

The main results are presented in Table 2. The absolute error rates on the test subset with the 1NN method are shown for each distance. We compare the new distance measures  $TD_{DTW}^C$ ,  $TD_{DTW}^S$ , and  $TD_{DTW}^H$  with the standard DTW,  $DD_{DTW}$  method published in [12], and transform components  $T_{DTW}^C$ ,  $T_{DTW}^S$ ,  $T_{DTW}^H$  of new distances.  $DD_{DTW}$  is similar to the distance in Eq. (1), where instead of the transform  $\hat{f}$  we take the first derivative  $f'$  of the function. The component methods  $T_{DTW}^C$ ,  $T_{DTW}^S$ ,  $T_{DTW}^H$  are distances, where we take only the DTW distance of the data transformed by transforms C, S, H respectively. This is the case of Eq. (1) with  $a = 0$ ,  $b = 1$ .

Comparing the new distances with the standard DTW we can see a significant reduction in error rate for most data sets. This is especially clearly seen for the mean of relative errors presented in Table 3. The average relative error reduction for all data sets is equal to 9.44% for  $TD_{DTW}^C$ , 8.24% for  $TD_{DTW}^S$ , and 5.96% for  $TD_{DTW}^H$ . The component distances  $T_{DTW}^C$ ,  $T_{DTW}^S$ ,  $T_{DTW}^H$  are much worse than DTW, as we could expect. However  $DD_{DTW}$  distance is still better than the new distances, with a 19.3% reduction in relative error rate compared with DTW. A graphical comparison of TD methods and DTW is presented in Fig. 3. We see that the new methods are clearly superior to DTW distance on most of the examined data sets.

An interesting question is how the distance on the function and the distance on the transform of the function contribute to the final distance. We can see this in Figs. 4 and 5. In Fig. 4 it is shown that there is no constant  $\alpha$  with best result of classification. For different data sets and different transforms we have different optimal values of the parameter  $\alpha$ . Similarly, for one fixed data set, the minimum of the error rates can be obtained for various values of  $\alpha$  (Fig. 5).

To find differences between the methods we present a detailed statistical comparison.

We test the null hypothesis that all classifiers perform the same and the observed differences are merely random. We used the Iman and Davenport [14] test, which is a non-parametric equivalent of ANOVA. Let  $R_{ij}$  be the rank of the  $j$ th of  $K$  classifiers on the  $i$ th of  $N$  data sets and  $R_j = \frac{1}{N} \sum_{i=1}^N R_{ij}$ . The test compares the mean ranks of classifiers and is based on the statistic

$$S' = \frac{(N-1)S}{N(K-1) - S}, \quad (2)$$

where

$$S = \frac{12N}{K(K+1)} \sum_{i=1}^K R_i^2 - 3N(K+1) \quad (3)$$

is the Friedman statistic, which has the  $F$  distribution with  $K-1$  and  $(K-1)(N-1)$  degrees of freedom.

If we consider the classification errors on all data sets, then in our case  $N = 47$  and  $K = 8$ . The value of the statistic  $S'$  is equal to 14.60, and the corresponding critical value is equal to  $F(0.95, 7, 322) = 2.04$ . Due to the fact that the critical value is lower than the respective statistic ( $p$ -value equal to 0), we can proceed with the post hoc tests in order to detect significant pairwise differences among all the classifiers. A set of pairwise comparisons can be associated with a set of hypotheses. Any of the post hoc tests which can be applied to non-parametric tests work over a family of hypotheses. The test statistic for comparing the  $i$ th and  $j$ th classifier is

$$Z = \frac{R_i - R_j}{\sqrt{\frac{K(K+1)}{6N}}}.$$

This statistic is asymptotically normal with zero mean and unit variance.

**Table 1**  
Summary of data sets.

| Data set                       | Number of classes | Size of training set | Size of testing set | Time series length |
|--------------------------------|-------------------|----------------------|---------------------|--------------------|
| 50Words                        | 50                | 450                  | 455                 | 270                |
| Adiac                          | 37                | 390                  | 391                 | 176                |
| Beef                           | 5                 | 30                   | 30                  | 470                |
| Car                            | 4                 | 60                   | 60                  | 577                |
| CBF                            | 3                 | 30                   | 900                 | 128                |
| ChlorineConcentration          | 3                 | 467                  | 3840                | 166                |
| CinC_ECG_torso                 | 4                 | 40                   | 1380                | 1639               |
| Coffee                         | 2                 | 28                   | 28                  | 286                |
| Cricket_X                      | 12                | 390                  | 390                 | 300                |
| Cricket_Y                      | 12                | 390                  | 390                 | 300                |
| Cricket_Z                      | 12                | 390                  | 390                 | 300                |
| DiatomSizeReduction            | 4                 | 16                   | 306                 | 345                |
| ECG200                         | 2                 | 100                  | 100                 | 96                 |
| ECGFiveDays                    | 2                 | 23                   | 861                 | 136                |
| Face (all)                     | 14                | 560                  | 1690                | 131                |
| Face (four)                    | 4                 | 24                   | 88                  | 350                |
| FacesUCR                       | 14                | 200                  | 2050                | 131                |
| Fish                           | 7                 | 175                  | 175                 | 463                |
| Gun-Point                      | 2                 | 50                   | 150                 | 150                |
| Haptics                        | 5                 | 155                  | 308                 | 1092               |
| InlineSkate                    | 7                 | 100                  | 550                 | 1882               |
| ItalyPowerDemand               | 2                 | 67                   | 1029                | 24                 |
| Lightning-2                    | 2                 | 60                   | 61                  | 637                |
| Lightning-7                    | 7                 | 70                   | 73                  | 319                |
| MALLAT                         | 8                 | 55                   | 2345                | 1024               |
| MedicalImages                  | 10                | 381                  | 760                 | 99                 |
| MoteStrain                     | 2                 | 20                   | 1252                | 84                 |
| Non-Invasive Fetal ECG Thorax1 | 42                | 1800                 | 1965                | 750                |
| Non-Invasive Fetal ECG Thorax2 | 42                | 1800                 | 1965                | 750                |
| OliveOil                       | 4                 | 30                   | 30                  | 570                |
| OSU Leaf                       | 6                 | 200                  | 242                 | 427                |
| Plane                          | 7                 | 105                  | 105                 | 144                |
| SonyAIBORobot Surface          | 2                 | 20                   | 601                 | 70                 |
| SonyAIBORobot SurfaceII        | 2                 | 27                   | 953                 | 65                 |
| StarLightCurves                | 3                 | 1000                 | 8236                | 1024               |
| Swedish Leaf                   | 15                | 500                  | 625                 | 128                |
| Symbols                        | 6                 | 25                   | 995                 | 398                |
| Synthetic Control              | 6                 | 300                  | 300                 | 60                 |
| Trace                          | 4                 | 100                  | 100                 | 275                |
| Two Patterns                   | 4                 | 1000                 | 4000                | 128                |
| TwoLeadECG                     | 2                 | 23                   | 1139                | 82                 |
| uWaveGestureLibrary_X          | 8                 | 896                  | 3582                | 315                |
| uWaveGestureLibrary_Y          | 8                 | 896                  | 3582                | 315                |
| uWaveGestureLibrary_Z          | 8                 | 896                  | 3582                | 315                |
| Wafer                          | 2                 | 1000                 | 6174                | 152                |
| WordsSynonyms                  | 25                | 267                  | 638                 | 270                |
| Yoga                           | 2                 | 300                  | 3000                | 426                |

When comparing multiple algorithms, to retain an overall significance level  $\alpha$ , one has to adjust the value of  $\alpha$  for each post hoc comparison. There are various methods for this. Garcia and Herrera [10] explain and compare the use of various correction algorithms. They showed that although it requires intensive computation, the Bergmann and Hommel [5] dynamic procedure has the highest power. This procedure is based on the idea of finding all elementary hypotheses which cannot be rejected. A rapid algorithm [13] allows a substantial reduction in computing time.

The results of multiple comparisons are given in Table 4. Those classifiers that are connected by a sequence of stars have average ranks that are not significantly different from one another. We have three homogeneous groups of classifiers.

#### 4. 3D methods

As we have seen in the previous section, adding information about the derivative or some transforms improves the classification process. It seems interesting to ask whether a combination of the two approaches leads to further decrease in the classification error. In this section we will address this problem, i.e. we will

combine information about a function (time series), its derivative and its transform in one distance measure.

##### 4.1. Distance measure

If  $\text{dist}$  is a distance measure for two time series  $f$  and  $g$ , a new distance measure  $\widehat{\text{dist}}_{abc}$  is defined by

$$\widehat{\text{dist}}_{abc}(f, g) := a \text{dist}(f, g) + b \text{dist}(f', g') + c \text{dist}(\hat{f}, \hat{g}), \quad (4)$$

where  $f'$ ,  $g'$  are discrete (first) derivatives of  $f$ ,  $g$ ;  $\hat{f}$ ,  $\hat{g}$  are transforms; and  $a, b, c \in [0, 1]$  are parameters. The discrete derivative of a time series  $f$  with length  $n$  is defined by

$$F'(i) = f(i) - f(i-1), \quad i = 2, 3, \dots, n$$

where  $f'$  is a time series with length  $n-1$ .

The parameters  $a$ ,  $b$ ,  $c$  are chosen in the tuning phase of a learning process. For an arbitrary distance function  $\text{dist}$ , we will denote the new distance measure by  $\text{DTD}_{\text{dist}}^{\text{trans}}$  (derivative-transform distance), for example  $\text{DTD}_{\text{DTW}}^{\text{C}}$  (for the cosine transform).



**Table 2**

Testing error rates.

|                                | DTW   | DD <sub>DTW</sub> | T <sub>DTW</sub> <sup>C</sup> | T <sub>DTW</sub> <sup>S</sup> | T <sub>DTW</sub> <sup>H</sup> | TD <sub>DTW</sub> <sup>C</sup> | TD <sub>DTW</sub> <sup>S</sup> | TD <sub>DTW</sub> <sup>H</sup> |
|--------------------------------|-------|-------------------|-------------------------------|-------------------------------|-------------------------------|--------------------------------|--------------------------------|--------------------------------|
| 50Words                        | 30.99 | 24.62             | 44.40                         | 41.98                         | 33.85                         | 30.55                          | 28.57                          | 29.01                          |
| Adiac                          | 39.64 | 29.92             | 38.62                         | 38.87                         | 44.50                         | 38.62                          | 38.36                          | 37.60                          |
| Beef                           | 50.00 | 43.33             | 53.33                         | 50.00                         | 46.67                         | 50.00                          | 50.00                          | 46.67                          |
| Car                            | 26.67 | 20.00             | 35.00                         | 28.33                         | 31.67                         | 26.67                          | 30.00                          | 26.67                          |
| CBF                            | 0.33  | 0.33              | 2.56                          | 5.89                          | 0.00                          | 0.33                           | 0.33                           | 0.33                           |
| ChlorineConcentration          | 35.16 | 29.19             | 35.16                         | 35.10                         | 37.94                         | 35.13                          | 35.05                          | 35.94                          |
| CinC_ECG_torso                 | 34.93 | 27.54             | 14.78                         | 15.43                         | 30.80                         | 20.00                          | 21.74                          | 30.36                          |
| Coffee                         | 17.86 | 10.71             | 28.57                         | 25.00                         | 21.43                         | 17.86                          | 17.86                          | 17.86                          |
| Cricket_X                      | 22.31 | 21.79             | 32.05                         | 33.85                         | 22.31                         | 22.31                          | 20.77                          | 21.54                          |
| Cricket_Y                      | 20.77 | 20.77             | 29.74                         | 30.00                         | 18.97                         | 20.77                          | 20.77                          | 17.44                          |
| Cricket_Z                      | 20.77 | 21.54             | 32.05                         | 31.54                         | 19.23                         | 19.23                          | 19.23                          | 17.44                          |
| DiatomSizeReduction            | 3.27  | 3.27              | 8.50                          | 6.21                          | 5.56                          | 3.27                           | 3.27                           | 3.27                           |
| ECG200                         | 23.00 | 17.00             | 16.00                         | 8.00                          | 19.00                         | 16.00                          | 9.00                           | 16.00                          |
| ECGFiveDays                    | 23.23 | 23.11             | 17.42                         | 21.25                         | 20.33                         | 16.96                          | 19.28                          | 21.14                          |
| Face (all)                     | 19.23 | 9.82              | 36.63                         | 28.93                         | 23.31                         | 19.47                          | 19.59                          | 20.30                          |
| Face (four)                    | 17.05 | 17.05             | 35.23                         | 36.36                         | 17.05                         | 17.05                          | 17.05                          | 14.77                          |
| FacesUCR                       | 9.51  | 9.56              | 28.44                         | 29.76                         | 8.20                          | 8.93                           | 9.66                           | 8.34                           |
| Fish                           | 16.57 | 5.71              | 26.29                         | 21.71                         | 31.43                         | 21.14                          | 17.71                          | 14.86                          |
| Gun-Point                      | 9.33  | 2.00              | 10.00                         | 8.00                          | 10.67                         | 6.67                           | 7.33                           | 9.33                           |
| Haptics                        | 62.34 | 60.06             | 62.01                         | 61.04                         | 59.09                         | 59.42                          | 59.09                          | 59.09                          |
| InlineSkate                    | 61.64 | 43.82             | 66.73                         | 68.55                         | 60.55                         | 61.64                          | 61.64                          | 60.00                          |
| ItalyPowerDemand               | 4.96  | 4.96              | 4.86                          | 5.25                          | 7.48                          | 4.96                           | 4.96                           | 4.96                           |
| Lighting-2                     | 13.11 | 13.11             | 16.39                         | 16.39                         | 9.84                          | 8.20                           | 9.84                           | 8.20                           |
| Lighting-7                     | 27.40 | 32.88             | 26.03                         | 27.40                         | 32.88                         | 27.40                          | 23.29                          | 27.40                          |
| MALLAT                         | 6.61  | 5.12              | 7.29                          | 7.97                          | 11.47                         | 4.90                           | 4.69                           | 6.65                           |
| MedicalImages                  | 26.32 | 26.32             | 27.24                         | 30.79                         | 28.68                         | 25.26                          | 25.26                          | 25.26                          |
| MoteStrain                     | 16.53 | 16.69             | 21.33                         | 23.24                         | 13.82                         | 16.53                          | 16.13                          | 15.73                          |
| Non-Invasive Fetal ECG Thorax1 | 20.97 | 19.39             | 16.08                         | 16.79                         | 19.39                         | 15.52                          | 16.34                          | 17.25                          |
| Non-Invasive Fetal ECG Thorax2 | 13.54 | 10.69             | 12.37                         | 12.11                         | 10.69                         | 11.55                          | 11.96                          | 12.62                          |
| OliveOil                       | 13.33 | 13.33             | 13.33                         | 13.33                         | 16.67                         | 13.33                          | 16.67                          | 16.67                          |
| OSU Leaf                       | 40.91 | 11.98             | 50.41                         | 52.48                         | 44.63                         | 38.02                          | 40.08                          | 43.80                          |
| Plane                          | 0.00  | 0.00              | 2.86                          | 3.81                          | 0.00                          | 0.00                           | 0.00                           | 0.00                           |
| SonyAIBORobot Surface          | 27.45 | 25.79             | 33.78                         | 37.44                         | 25.96                         | 31.61                          | 27.45                          | 27.95                          |
| SonyAIBORobot SurfaceII        | 16.89 | 10.81             | 19.41                         | 16.89                         | 20.25                         | 18.15                          | 16.89                          | 18.47                          |
| StarLightCurves                | 9.34  | 3.78              | 10.27                         | 12.08                         | 9.74                          | 9.62                           | 9.56                           | 8.50                           |
| Swedish Leaf                   | 20.80 | 9.92              | 22.88                         | 22.56                         | 19.52                         | 12.80                          | 12.32                          | 15.04                          |
| Symbols                        | 5.03  | 4.72              | 23.72                         | 23.12                         | 3.12                          | 5.03                           | 5.03                           | 4.12                           |
| Synthetic Control              | 0.67  | 0.67              | 6.33                          | 4.33                          | 3.33                          | 0.33                           | 0.67                           | 0.67                           |
| Trace                          | 0.00  | 0.00              | 9.00                          | 16.00                         | 0.00                          | 0.00                           | 0.00                           | 0.00                           |
| Two Patterns                   | 0.00  | 0.00              | 40.55                         | 37.90                         | 0.00                          | 0.00                           | 0.00                           | 0.00                           |
| TwoLeadECG                     | 9.57  | 2.19              | 28.53                         | 28.80                         | 16.24                         | 9.57                           | 9.57                           | 9.57                           |
| uWaveGestureLibrary_X          | 27.25 | 22.08             | 39.06                         | 31.77                         | 25.68                         | 23.73                          | 24.90                          | 23.95                          |
| uWaveGestureLibrary_Y          | 36.60 | 28.45             | 43.44                         | 40.31                         | 37.16                         | 30.68                          | 31.18                          | 33.33                          |
| uWaveGestureLibrary_Z          | 34.17 | 30.40             | 40.73                         | 38.67                         | 35.57                         | 31.55                          | 33.14                          | 32.55                          |
| Wafer                          | 2.01  | 2.01              | 0.70                          | 0.83                          | 2.35                          | 0.75                           | 0.94                           | 2.04                           |
| WordsSynonyms                  | 35.11 | 26.96             | 47.81                         | 44.98                         | 32.76                         | 28.06                          | 30.56                          | 30.41                          |
| Yoga                           | 16.37 | 14.40             | 18.13                         | 17.73                         | 17.33                         | 17.10                          | 16.37                          | 16.23                          |

**Table 3**

Average relative testing error rates on all data sets.

| $\frac{DD_{DTW}-DTW}{DTW}$ | $\frac{T_{DTW}^C-DTW}{DTW}$ | $\frac{T_{DTW}^S-DTW}{DTW}$ | $\frac{T_{DTW}^H-DTW}{DTW}$ | $\frac{TD_{DTW}^C-DTW}{DTW}$ | $\frac{TD_{DTW}^S-DTW}{DTW}$ | $\frac{TD_{DTW}^H-DTW}{DTW}$ |
|----------------------------|-----------------------------|-----------------------------|-----------------------------|------------------------------|------------------------------|------------------------------|
| -19.03                     | 69.46                       | 81.07                       | 14.91                       | -9.44                        | -8.24                        | -5.96                        |

#### 4.2. Parameter dimension

We do not have to check all values of  $a, b, c \in [0, 1]$ . If  $a_1 = ka_2$ ,  $b_1 = kb_2$ ,  $c_1 = kc_2$  where  $k > 0$  is a constant (i.e. points  $(a_1, b_1, c_1)$ ,  $(a_2, b_2, c_2)$  are linearly dependent), we have

$$\widehat{\text{dist}}_{a_1 b_1 c_1}(f_1, g_1) \leq \widehat{\text{dist}}_{a_1 b_1 c_1}(f_2, g_2) \iff \widehat{\text{dist}}_{a_2 b_2 c_2}(f_1, g_1) \leq \widehat{\text{dist}}_{a_2 b_2 c_2}(f_2, g_2)$$

hence we can choose points  $(a, b, c)$  on any continuous surface between the points  $A = (1, 0, 0)$ ,  $B = (0, 1, 0)$ , and  $C = (0, 0, 1)$ . For example, this may be the area of the triangle with vertices at those points, or one-eighth of a sphere. For simplicity, we choose the triangle. This 3D triangle is mapped to the 2D triangle with vertices at the points  $A' = (0, 0)$ ,  $B' = (1, 0)$ , and  $C' = (\frac{1}{2}, \frac{\sqrt{3}}{2})$ . Both triangles can be defined in parametrical fashion:

$$(a, b, c) = A + \alpha \overrightarrow{AB} + \beta \overrightarrow{AC},$$

$$(a', b') = A' + \alpha \overrightarrow{A'B'} + \beta \overrightarrow{A'C'},$$

where  $(a, b, c)$  are points of the 3D triangle,  $(a', b')$  are points of the 2D triangle, and  $\alpha \in [0, 1]$ ,  $\beta \in [0, 1 - \alpha]$  are parameters (Fig. 6).

In what follows we will use the 2D parameters  $\alpha$ ,  $\beta$  (or  $a', b'$ ) instead of the 3D parameters  $a$ ,  $b$ ,  $c$ .

#### 4.3. Lower bound and triangle inequality

If low is a lower bound for a distance dist, then

$$\widehat{\text{low}}_{abc}(f, g) = a \text{low}(f, g) + b \text{low}(f', g') + c \text{low}(f, \hat{g})$$

is a lower bound for the distance  $\widehat{\text{dist}}_{abc}$ .

If the base distance dist is a metric, then the new distance  $\widehat{\text{dist}}$  is also a metric. If dist is not a metric but obeys the triangle inequality, then the distance  $\widehat{\text{dist}}$  obeys the triangle inequality as well:

$$\widehat{\text{dist}}_{abc}(f, g) \leq \widehat{\text{dist}}_{abc}(f, h) + \widehat{\text{dist}}_{abc}(h, g).$$

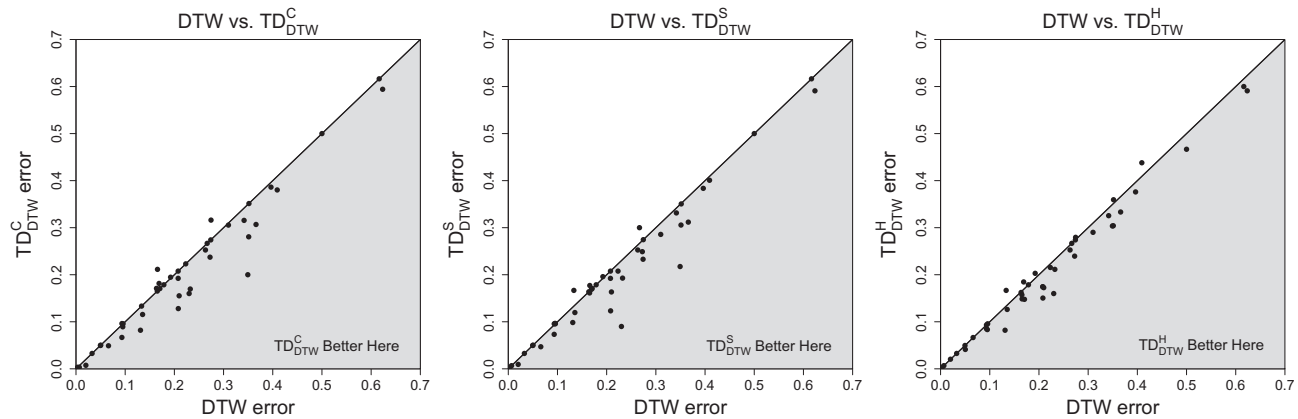


Fig. 3. Comparison of test errors (DTW vs.  $TD^i_{DTW}$ ).

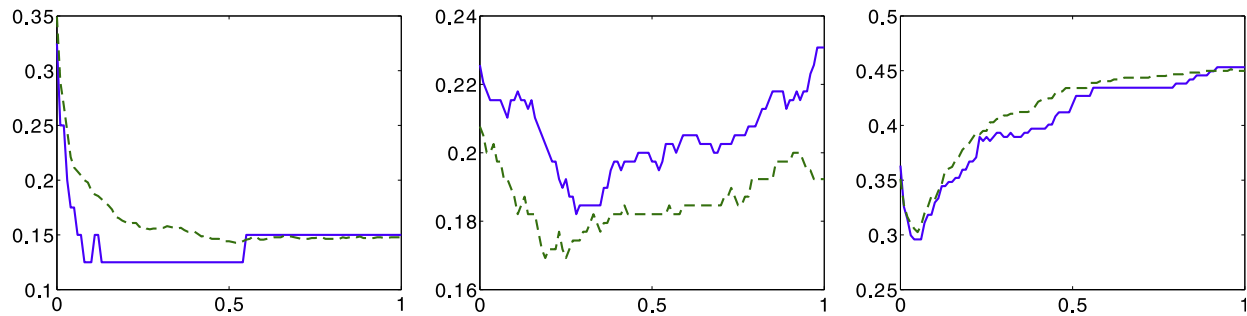


Fig. 4. Dependence of classification error on the participation of the transforms (parameter  $\alpha$ ) for TD methods on example data sets. From left:  $TD^C_{DTW}$  on *CinC-ECG-torse*,  $TD^H_{DTW}$  on *Cricket-Z*,  $TD^S_{DTW}$  on *WordsSynonyms* (dashed line CV error, solid line test error).

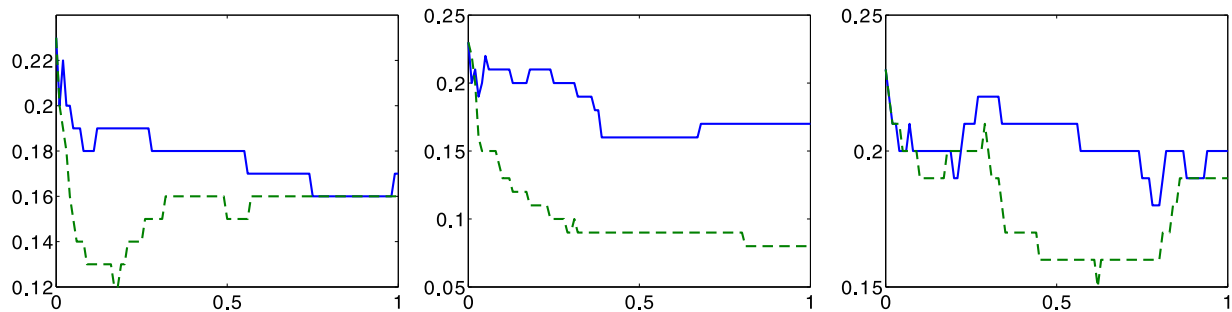


Fig. 5. Dependence of classification error on the participation of the transforms (parameter  $\alpha$ ) on the same *ecg200* data set. From left:  $TD^C_{DTW}$ ,  $TD^S_{DTW}$ ,  $TD^H_{DTW}$  (dashed line CV error, solid line test error).

Table 4  
Results of the Bergmann–Hommel post hoc test.

| Procedure    | Ranks mean |   |   |   |
|--------------|------------|---|---|---|
| $DD_{DTW}$   | 3.16       | * |   |   |
| $TD^S_{DTW}$ | 3.54       | * | * |   |
| $TD^H_{DTW}$ | 3.55       | * | * |   |
| $TD^C_{DTW}$ | 3.71       | * | * |   |
| DTW          | 4.06       |   | * |   |
| $T^H_{DTW}$  | 5.10       |   | * | * |
| $T^C_{DTW}$  | 6.14       |   |   | * |
| $T^S_{DTW}$  | 6.18       |   |   | * |

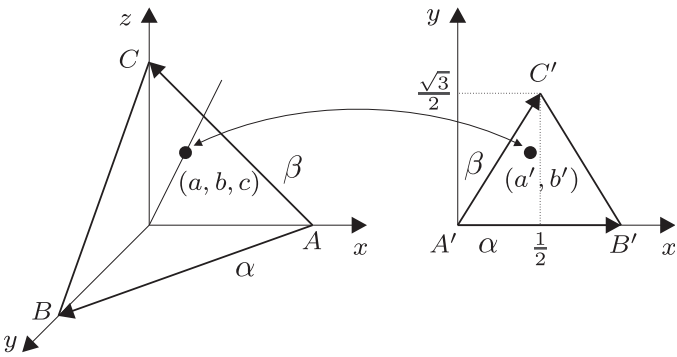


Fig. 6. Relationship between the 3D parameters  $a, b, c$  and 2D parameters  $\alpha, \beta$ .

**Table 5**

Testing error rates.

|                                | DTW   | DD <sub>DTW</sub> | DTD <sub>DTW</sub> <sup>C</sup> | DTD <sub>DTW</sub> <sup>S</sup> | DTD <sub>DTW</sub> <sup>H</sup> |
|--------------------------------|-------|-------------------|---------------------------------|---------------------------------|---------------------------------|
| 50Words                        | 30.99 | 24.62             | 24.62                           | 24.62                           | 25.05                           |
| Adiac                          | 39.64 | 29.92             | 29.92                           | 29.92                           | 30.95                           |
| Beef                           | 50.00 | 43.33             | 43.33                           | 43.33                           | 46.67                           |
| Car                            | 26.67 | 20.00             | 20.00                           | 30.00                           | 25.00                           |
| CBF                            | 0.33  | 0.33              | 0.33                            | 0.33                            | 0.33                            |
| ChlorineConcentration          | 35.16 | 29.19             | 29.19                           | 29.19                           | 29.32                           |
| CinC_ECG_torso                 | 34.93 | 27.54             | 17.75                           | 17.46                           | 29.78                           |
| Coffee                         | 17.86 | 10.71             | 10.71                           | 10.71                           | 10.71                           |
| Cricket_X                      | 22.31 | 21.79             | 22.05                           | 21.79                           | 21.28                           |
| Cricket_Y                      | 20.77 | 20.77             | 20.77                           | 19.74                           | 17.18                           |
| Cricket_Z                      | 20.77 | 21.54             | 19.74                           | 19.23                           | 17.44                           |
| DiatomSizeReduction            | 3.27  | 3.27              | 3.27                            | 3.27                            | 3.27                            |
| ECG200                         | 23.00 | 17.00             | 19.00                           | 14.00                           | 16.00                           |
| ECGFiveDays                    | 23.23 | 23.11             | 18.23                           | 19.40                           | 20.09                           |
| Face (all)                     | 19.23 | 9.82              | 10.89                           | 10.65                           | 20.30                           |
| Face (four)                    | 17.05 | 17.05             | 17.05                           | 17.05                           | 14.77                           |
| FacesUCR                       | 9.51  | 9.56              | 9.07                            | 9.22                            | 7.80                            |
| Fish                           | 16.57 | 5.71              | 5.71                            | 5.71                            | 6.29                            |
| Gun-Point                      | 9.33  | 2.00              | 2.00                            | 2.00                            | 1.33                            |
| Haptics                        | 62.34 | 60.06             | 55.84                           | 58.77                           | 57.14                           |
| InlineSkate                    | 61.64 | 43.82             | 43.82                           | 43.82                           | 53.27                           |
| ItalyPowerDemand               | 4.96  | 4.96              | 4.96                            | 4.96                            | 4.96                            |
| Lighting-2                     | 13.11 | 13.11             | 8.20                            | 9.84                            | 8.20                            |
| Lighting-7                     | 27.40 | 32.88             | 23.29                           | 20.55                           | 27.40                           |
| MALLAT                         | 6.61  | 5.12              | 5.12                            | 5.12                            | 6.65                            |
| MedicalImages                  | 26.32 | 26.32             | 25.66                           | 27.50                           | 25.66                           |
| MoteStrain                     | 16.53 | 16.69             | 19.33                           | 16.69                           | 16.69                           |
| Non-Invasive Fetal ECG Thorax1 | 20.97 | 19.39             | 15.22                           | 15.67                           | 15.22                           |
| Non-Invasive Fetal ECG Thorax2 | 13.54 | 10.69             | 10.59                           | 10.89                           | 11.30                           |
| OliveOil                       | 13.33 | 13.33             | 13.33                           | 13.33                           | 16.67                           |
| OSU Leaf                       | 40.91 | 11.98             | 11.98                           | 11.98                           | 11.98                           |
| Plane                          | 0.00  | 0.00              | 0.00                            | 0.00                            | 0.00                            |
| SonyAIBORobot Surface          | 27.45 | 25.79             | 25.79                           | 25.79                           | 26.62                           |
| SonyAIBORobot SurfaceII        | 16.89 | 10.81             | 10.81                           | 10.81                           | 10.49                           |
| StarLightCurves                | 9.34  | 3.78              | 3.78                            | 3.78                            | 3.80                            |
| Swedish Leaf                   | 20.80 | 9.92              | 8.32                            | 9.12                            | 9.92                            |
| Symbols                        | 5.03  | 4.72              | 4.72                            | 4.72                            | 4.12                            |
| Synthetic Control              | 0.67  | 0.67              | 0.33                            | 0.67                            | 0.67                            |
| Trace                          | 0.00  | 0.00              | 0.00                            | 0.00                            | 0.00                            |
| Two Patterns                   | 0.00  | 0.00              | 0.00                            | 0.00                            | 0.00                            |
| TwoLeadECG                     | 9.57  | 2.19              | 2.19                            | 2.19                            | 2.19                            |
| uWaveGestureLibrary_X          | 27.25 | 22.08             | 21.94                           | 22.72                           | 21.02                           |
| uWaveGestureLibrary_Y          | 36.60 | 28.45             | 28.45                           | 28.45                           | 28.11                           |
| uWaveGestureLibrary_Z          | 34.17 | 30.40             | 30.49                           | 30.82                           | 30.07                           |
| Wafer                          | 2.01  | 2.01              | 0.75                            | 0.94                            | 2.04                            |
| WordsSynonyms                  | 35.11 | 26.96             | 26.49                           | 26.80                           | 26.49                           |
| Yoga                           | 16.37 | 14.40             | 14.40                           | 14.40                           | 14.77                           |

#### 4.4. Experimental setup

We performed experiments on the 47 data sets listed in Table 1. For use with the new distance function, we chose one distance measure, DTW, and three transforms: sine, cosine, and Hilbert. Thus we have three similarity measures denoted by  $DTD_{DTW}^S$ ,  $DTD_{DTW}^C$ , and  $DTD_{DTW}^H$ . For each data set we calculated the classification error rate on a test subset (to learn the model we used a training subset with the leave-one-out, 1NN method). We found all parameters using the training subset.

We use the cross-validation (leave-one-out) method to find the best pair of parameters  $\alpha$ ,  $\beta$  in our classifier. If the minimal error rate is the same for more than one pair of parameters  $\alpha$ ,  $\beta$ , we choose the smallest pair – minimizing  $\beta$  first, then  $\alpha$ . As a result of this method of minimization, if  $DTD_{DTW}^*$  and  $DD_{DTW}$  have the same CV error rate, then they have the same test error rate as well. We implemented the algorithm discussed in Section 3.3. Finite subsets of parameters  $\alpha$  and  $\beta$  are chosen, from 0 to 1 with fixed step 0.01. The code of the algorithm is similar to the code in Section 3.5. All that differs is the method of computation of param-

**Table 6**

Average relative testing error rates on all data sets.

| $\frac{DD_{DTW}-DTW}{DTW}$ | $\frac{DTD_{DTW}^C-DTW}{DTW}$ | $\frac{DTD_{DTW}^S-DTW}{DTW}$ | $\frac{DTD_{DTW}^H-DTW}{DTW}$ |
|----------------------------|-------------------------------|-------------------------------|-------------------------------|
| −19.03                     | −24.85                        | −22.97                        | −19.75                        |

ters  $a$ ,  $b$ ,  $c$  from  $\alpha$ ,  $\beta$  and the main line with computation of the distance value:

```
% e - list of time series in learning data set (cell
      vector of vectors)
% a, b, c - parameters computed from alpha, beta
% dist - distance function (DTW)
% diff - discrete (first) derivative
% trans - transform function (sine, cosine,
      Hilbert)
...
d = a * dist(e{j}, e{i})
    + b * dist(diff(e{j}), diff(e{i}))
    + c * dist(trans(e{j}), trans(e{i}));
...
```



#### 4.5. Results

The results are presented in Table 5. The absolute error rates on the test subset with the 1NN method are shown for each distance. We compare the new distance measures  $DTD_{DTW}^C$ ,  $DTD_{DTW}^S$ , and  $DTD_{DTW}^H$  with the standard DTW and  $DD_{DTW}$ . Comparing the new distances with the standard DTW we can see a significant reduction in error rate for most data sets. This is especially clearly seen for the mean of relative errors given in Table 6. The average relative error reduction for all data sets is equal to 24.85% for  $DTD_{DTW}^C$ , 22.97% for  $DTD_{DTW}^S$ , and 19.75% for  $TD_{DTW}^H$ . The reduction for the  $DD_{DTW}$  method is 19.03%, and therefore the new distances are slightly better than  $DD_{DTW}$ .

Graphical comparisons of DTD methods with DTW and of DTD with  $DD_{DTW}$  are presented in Figs. 7 and 8 respectively. We see that the new methods are clearly superior to the DTW and  $DD_{DTW}$  distances on most of the examined data sets.

It is not very useful to have a method that will perform well only on some problems, unless we can tell in advance on which problems it will be better. We must prove that we can predict ahead of time when our method will have better performance. Batista et al. [1,2] proposed a way of doing this. We can examine the accuracy of the compared methods looking only at the training data set, and use these results to choose an algorithm to use to classify the testing data set. We measure the accuracy of the examined methods on the training data set using leaving-one-out cross-validation, and calculate the expected change in accuracy as

$$\frac{\text{accuracy } DTD_{DTW}^*}{\text{accuracy } DD_{DTW}}.$$

Values greater than one indicate that we expect  $DTD_{DTW}^*$  to outperform  $DD_{DTW}$  on the given data set. In the same way we measure the actual change in accuracy using the testing data set. The results are shown in Fig. 9.

We can mark regions of the diagrams with the familiar labels:

- TP** In this region we correctly predicted that  $DTD_{DTW}^*$  would improve accuracy.
- TN** In this region we correctly predicted that  $DTD_{DTW}^*$  would decrease accuracy.
- FN** In this region we incorrectly predicted that  $DTD_{DTW}^*$  would decrease accuracy.
- FP** This region is the only truly bad case. Data points falling in this region represent data sets where we thought we could improve accuracy, but in fact did not.

Because  $DTD_{DTW}$  is an extension of  $DD_{DTW}$ , the error rate on the training data set is always no worse than the error rate for  $DD_{DTW}$ . Therefore the left part of Fig. 9 is empty. The vast majority of points lie in the TP area (especially for  $DTD_{DTW}^C$  and  $DTD_{DTW}^S$ ). This means that we are able to predict quite well when the method can improve the quality of classification. Of course, there are data sets for which the methods behave similarly: these lie near the point (1,1). In general, cases near the point (1,1) are not of special interest, since they represent marginal change in accuracy.

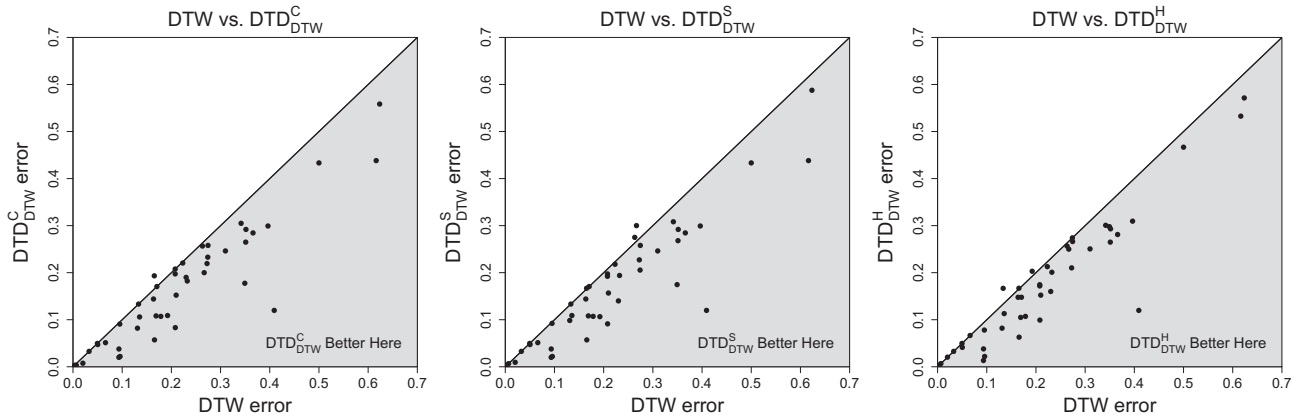


Fig. 7. Comparison of test errors (DTW vs.  $DTD_{DTW}^*$ ).

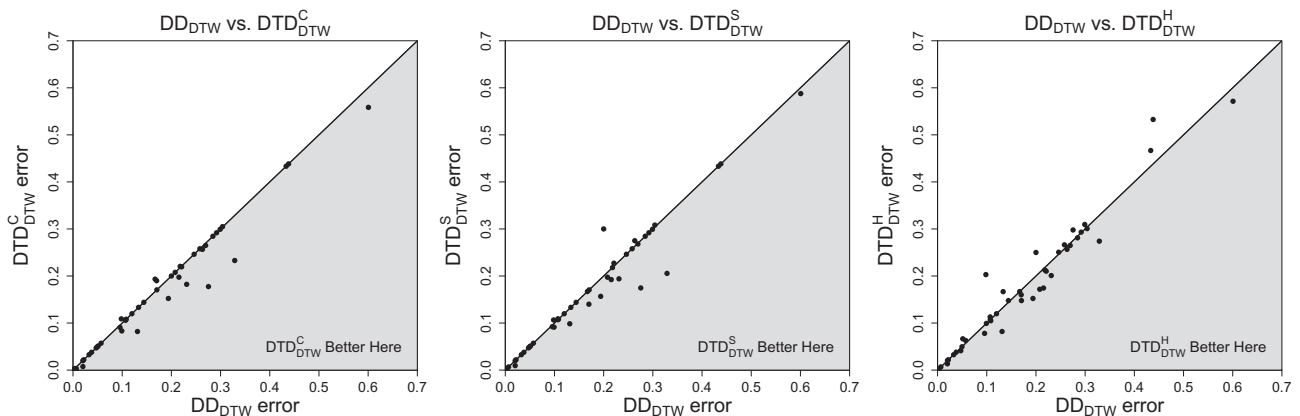


Fig. 8. Comparison of test errors ( $DD_{DTW}$  vs.  $DTD_{DTW}^*$ ).

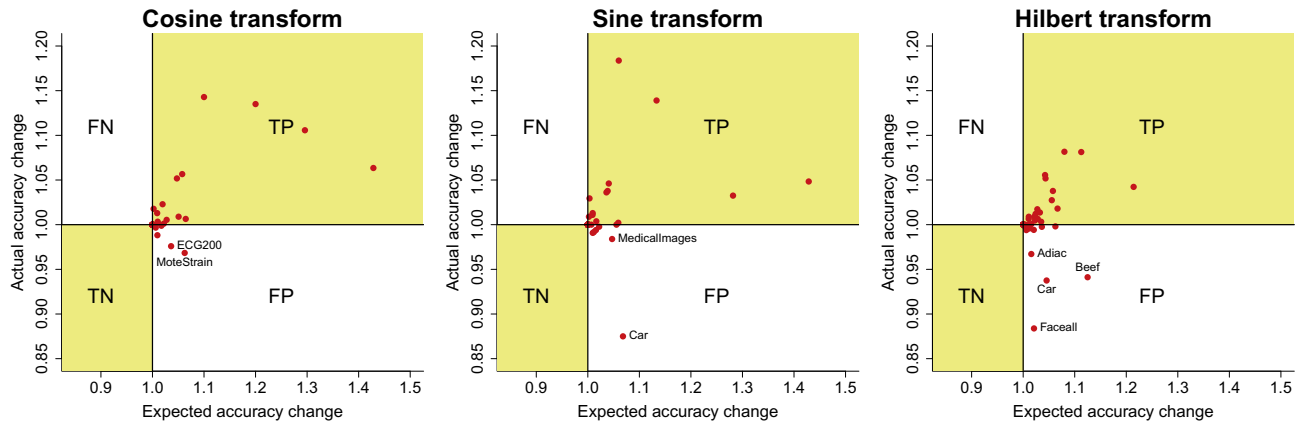


Fig. 9. Expected change in accuracy calculated on a training data set versus actual change in accuracy on a testing data set.

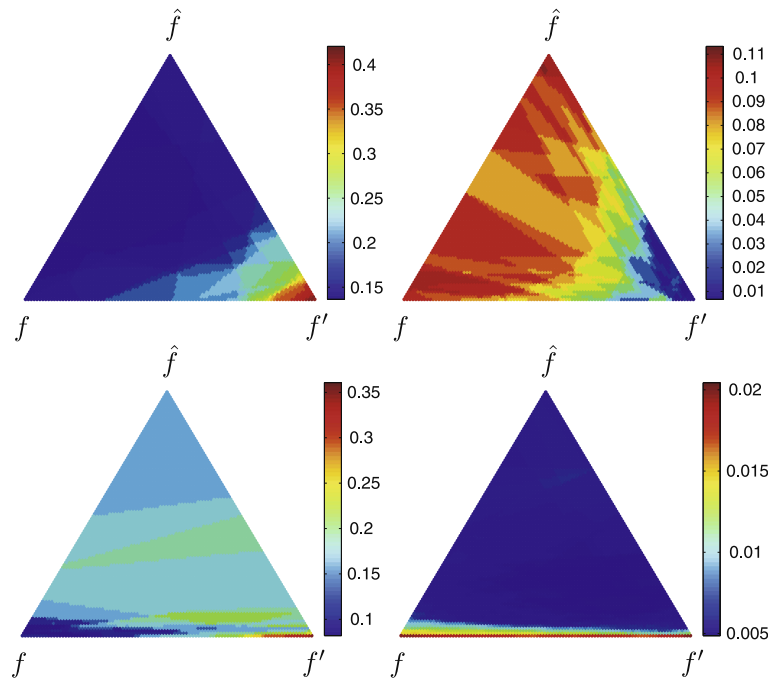


Fig. 10. Dependence of classification test error on the participation of the function  $f$ , its derivative  $f'$ , and its transform  $\hat{f}$  for DTD methods on example data sets. From top left:  $DTD_{DTW}^H$  on *facefour*,  $DTD_{DTW}^H$  on *gun-point*,  $DTD_{DTW}^S$  on *lighting2*,  $DTD_{DTW}^C$  on *wafer*.

Fig. 10 presents the contribution of each component of the new distance, i.e. the value of  $f$ , its derivative  $f'$  and its transform  $\hat{f}$ , for the example data sets. We can see that the minimum error can be obtained at different points of the triangle, i.e. for different values of the parameters  $a$ ,  $b$ ,  $c$  ( $\alpha, \beta$ ).

We can now use the Friedman and Bergmann–Hommel tests (described above) for statistical evaluation of the results. If we consider the classification errors on all data sets, then in our case  $N = 47$  and  $K = 5$ . The value of the statistic  $S'$  is equal to 14.91, and the corresponding critical value is equal to  $F(0.95, 4, 184) = 2.42$ . Due to the fact that the critical value is lower than the respective statistic (the  $p$ -value is equal to  $1.4E-10$ ), we can proceed with the post hoc test in order to detect significant pairwise differences among all the classifiers. The results of multiple comparisons are given in Table 7 and Table 8.

Those classifiers that are connected by a sequence of stars have average ranks that are not significantly different from one another.

Table 7

$p$ -values in the Bergmann–Hommel post hoc test.

| i  | Hypothesis                      | $p$ -Value           |
|----|---------------------------------|----------------------|
| 1  | DTW vs. $DTD_{DTW}^C$           | $1.6 \times 10^{-8}$ |
| 2  | DTW vs. $DTD_{DTW}^S$           | $4.4 \times 10^{-7}$ |
| 3  | DTW vs. $DTD_{DTW}^H$           | $4.0 \times 10^{-6}$ |
| 4  | DTW vs. $DD_{DTW}$              | 0.001                |
| 5  | $DD_{DTW}$ vs. $DTD_{DTW}^C$    | 0.087                |
| 6  | $DD_{DTW}$ vs. $DTD_{DTW}^S$    | 0.218                |
| 7  | $DD_{DTW}$ vs. $DTD_{DTW}^H$    | 0.384                |
| 8  | $DTD_{DTW}^C$ vs. $DTD_{DTW}^H$ | 0.761                |
| 9  | $DTD_{DTW}^C$ vs. $DTD_{DTW}^S$ | 0.761                |
| 10 | $DTD_{DTW}^S$ vs. $DTD_{DTW}^H$ | 0.761                |

Finally we obtained three homogeneous groups of classifiers. The best classifiers are in the first group. We can see that  $DTD_{DTW}^C$  is significantly better (10% level) than  $DD_{DTW}$ .

**Table 8**

Results of the Bergmann–Hommel post hoc test.

| Procedure     | Ranks mean |   |   |   |
|---------------|------------|---|---|---|
| $DTD_{DTW}^C$ | 2.33       | * |   |   |
| $DTD_{DTW}^S$ | 2.54       | * | * |   |
| $DTD_{DTW}^H$ | 2.70       | * | * |   |
| $DD_{DTW}$    | 3.13       |   | * |   |
| DTW           | 4.30       |   |   | * |

## 5. Conclusions and future work

In this paper we have introduced and studied an improvement on our previous time series distance based on derivatives. As it turns out, the addition of the transform gives very good results. The new classification method, which combines three distances: DTW distance between time series, DTW distance between derivatives of time series and DTW distance between transforms of time series, is superior to  $DD_{DTW}$ , which uses only the first two distances. The method with cosine transform can be especially recommended, as it is significantly better than  $DD_{DTW}$ . The proposed method, thanks to its parametrical approach, makes it possible to choose an appropriate model for any data set. Due to the high degree of nonlinearity, the method does not easily admit a rigorous theoretical analysis. However, the experiments that we have conducted provide evidence of the power and usefulness of the method.

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